OOP project documentation

(library system)

Executive summary about the project:

A library system project is software designed to manage and streamline the operations of a library. It typically includes the following key features:

1. **Catalog Management**: Helps organize and maintain information about books, journals, and other resources, including titles, authors, publication details, and genres.
2. **User Management**: Allows the library to register and manage members, track their profiles, and maintain borrowing history.
3. **Borrowing and Returns**: Tracks the lending and returning of books

4. **Search and Reservation**: Provides a searchable interface for users to locate books and reserve them.
5. **Inventory Management**: Helps the library maintain accurate records of available, borrowed, or missing resources.

Such a project can be implemented for public, academic, or private libraries, enhancing their efficiency and user experience.

Classes summary

User manger class:

- **A) Methods:**
  - **addcustomer(Library library, Scanner scanner): Prompts the user for customer details (username, name, password), creates a new Customer object, and saves it using the Writer class.**

- findCustomerById(String id): Searches for a customer by their ID by checking against a list retrieved from Reader.showallcustomers(). Returns the matching Customer object or null if not found.
- viewcustomers(Library library): Displays all registered customers. If no customers are found, it prints a corresponding message.
- addratingandreview(Book book, Scanner scanner): Prompts the user to rate a book and provide a review. If the user chooses to do so, it saves the rating and review using the Writer class.

Reader class:

A) Attributes:

- Several constant strings representing the filenames for various data files (books, customers, borrowed books, borrowhistory, *buyhistory)*.

- B) Methods:

- **readBooks(): Reads all books from the books.txt file and constructs a list of Book objects.**
- **readAvailableBooks(): Reads books from the same file but only returns those that are marked as "available".**
- **viewborrowedbooks(): Reads and returns a list of borrowed books from the borrowed.txt file.**
- **viewbought(): Reads and returns a list of purchased books from the buy history.txt file.**
- **viewborrowhistory(String customerid): Returns a list of books borrowed by a specific customer based on their ID from the borrow history.txt file.**
- **showallcustomers(): Reads customer data from the customer.txt file and returns a list of Customer objects.**
- **searchforbookid(String bookid): Searches for a specific book by its ID in the books.txt**

file and returns the corresponding Book object if found.

- ## Writer class:
- **A) Attributes:**
- **Several constant strings representing filenames for books, customers, boughthistory and borrowhistory.**

- **B) Methods:**
  - **removeBook(String bookid): Removes a book with the specified ID from the books.txt file and updates the file.**
  - **addbook(Book book): Appends a new book's details to the books.txt file.**
  - **addnewcustomer(Customer customer): Appends a new customer's details to the customer.txt file.**
  - **borrowbook(Book book, String customerID): Records the borrowing of a book by a customer in the borrowed.txt file.**

- returnbook(String bookId): Removes a book entry from the borrowed.txt file when it is returned.
- borrowhistory(Book book, String customerid): Records the borrowing history of a book for a customer in the borrow history.txt file.
- changeBookStatus(String bookId, String newStatus): Updates the status of a book (e.g., available, borrowed) in the books.txt file.
- boughthistory(Book book, String customerid): Records the purchase of a book in the buy history.txt file.
- saveRatingAndReview(Book book, double userRating, String review): Updates a book's rating and adds a new review in the books.txt file.

The customer class:

A) Attributes:

- id: Unique identifier for the customer.

- **name: Customer's name.**
- **password: Customer's password.**
- **cart: A list of books currently in the customer's shopping cart.**
- **borrowedBooks: A list of books currently borrowed by the customer.**
- **orderHistory: A list of strings representing the customer's past orders.**

B) Constructors:

- **A parameterized constructor to initialize the customer's attributes.**

C) Getters

1. **getId(): Returns the customer's unique identifier.**
2. **getName(): Returns the customer's name.**
3. **getPassword(): Returns the customer's password.**
4. **getCart(): Returns the list of books in the customer's shopping cart.**

5. **getBorrowedBooks(): Returns the list of books borrowed by the customer.**
6. **getOrderHistory(): Returns the list of order history details.**

D) Setters

1. **setId(String id): Sets the customer's unique identifier.**
2. **setName(String name): Sets the customer's name.**
3. **setPassword(String password): Sets the customer's password.**
4. **setCart(List<Book> cart): Sets the customer's shopping cart to the provided list of books.**
5. **setBorrowedBooks(List<Book> borrowedBooks): Sets the list of borrowed books to the provided list.**
6. **setOrderHistory(List<String> orderHistory): Sets the order history to the provided list of strings.**

- **E) Methods:**

- **validatePassword(String inputPassword): Checks if the provided password matches the customer's password.**
- **borrowBook(Book book): Adds a book to the borrowedBooks list.**
- **addOrder(String orderDetails): Records an order detail in the orderHistory.**

toString(): Returns a string representation of the

library system class (Main) :

A) **User Registration and Login**:

- Users can register and log in. Admin credentials are hardcoded for access to admin functionalities.

B) **Admin Functionalities**:

- Upon successful login, admins can add new books, view all books, remove books, and manage borrowed books and customer information.

C) **Customer Functionalities**:

- Customers can view available books, borrow books, return books, and purchase books. They also have the option to rate and review books after purchasing.

D) **Payment Methods**:

- Both cash and Visa are accepted for purchases and borrowings, with PIN validation for Visa transactions.

E) **Error Handling**:

- The system includes input validation and error messages for invalid choices and credentials.

F) **Program Flow**:

- The program runs in a loop allowing users to navigate through menus until they choose to exit.

Input and output scenarios

Main page:

1-register:asks you to enter a username/name/password the your user will be added

2-login:asks you to enter the username and the password you registered

3-exit: exit the entire system

Login page (user) :

1-view available books: show you the available books entered by the admin

2-borrow book: show you the available books and allow you to borrow

3-return book: let you return the book you borrowed

4-buy book: show you the available books and allow you to buy

5-Logout: send you to the main menu

Login page (admin) :

1-add book: allow admin to add a new book

2-view books: view the books added by the admin

3-remove book: remove the books that may not be available

4-view borrowed books: view the books borrowed by the users

5-view borrowed history: view the history of the users

6-view all customers: view customers that borrowed or bought books

8- back to main menu: send you to the main menu